

EXPERIMENT-1

Aim: Malware Detection via VirusTotal: Analyze a suspicious file/hash

Material Required:

1. Internet-connected computer
2. VirusTotal website: <https://www.virustotal.com>

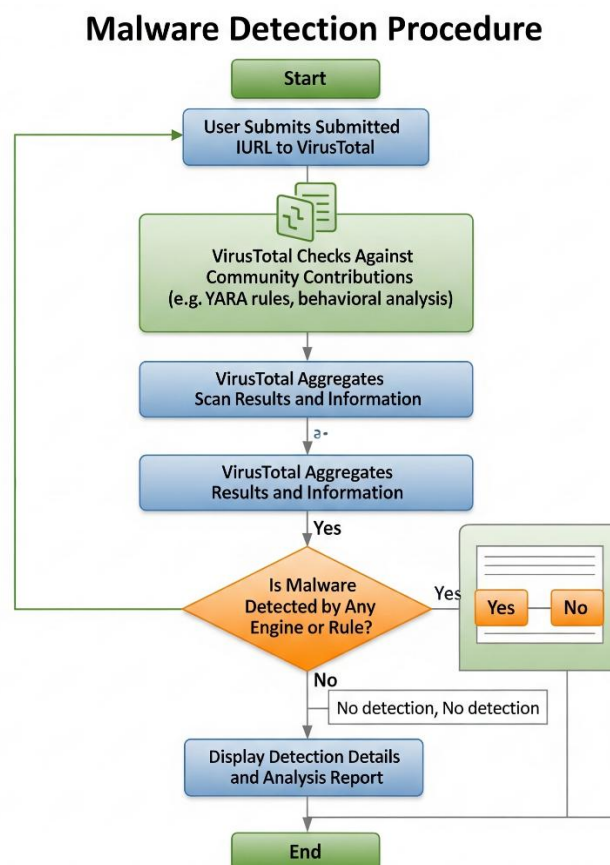
Theory:

VirusTotal is a free online tool used to scan and assess potentially dangerous files or links through various antivirus engines. It helps identify malware and provides detailed threat reports. Each security vendor uses its own detection systems. Domains behaving suspiciously—like hosting malware or phishing pages—can be flagged. A domain flagged by multiple sources is more likely to be dangerous.

Typical signs of malicious domains include:

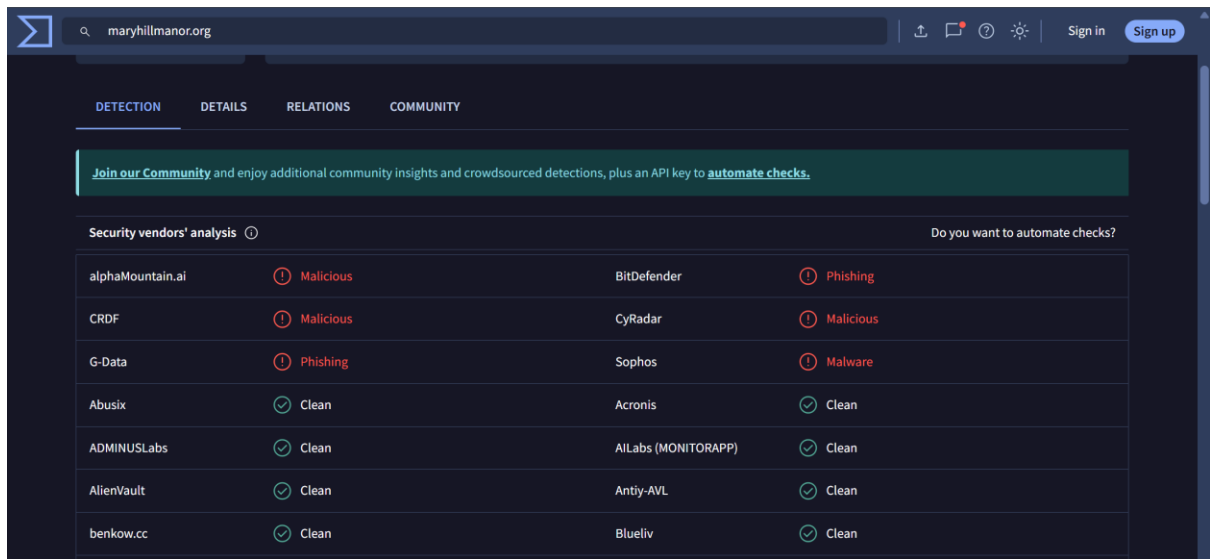
1. Newly registered domains with hidden ownership (WHOIS) details
2. Hosted in high-risk countries or IPs
3. Listed on known threat databases
4. Redirects to malicious downloads or sites

Procedure:



OBSERVATIONS

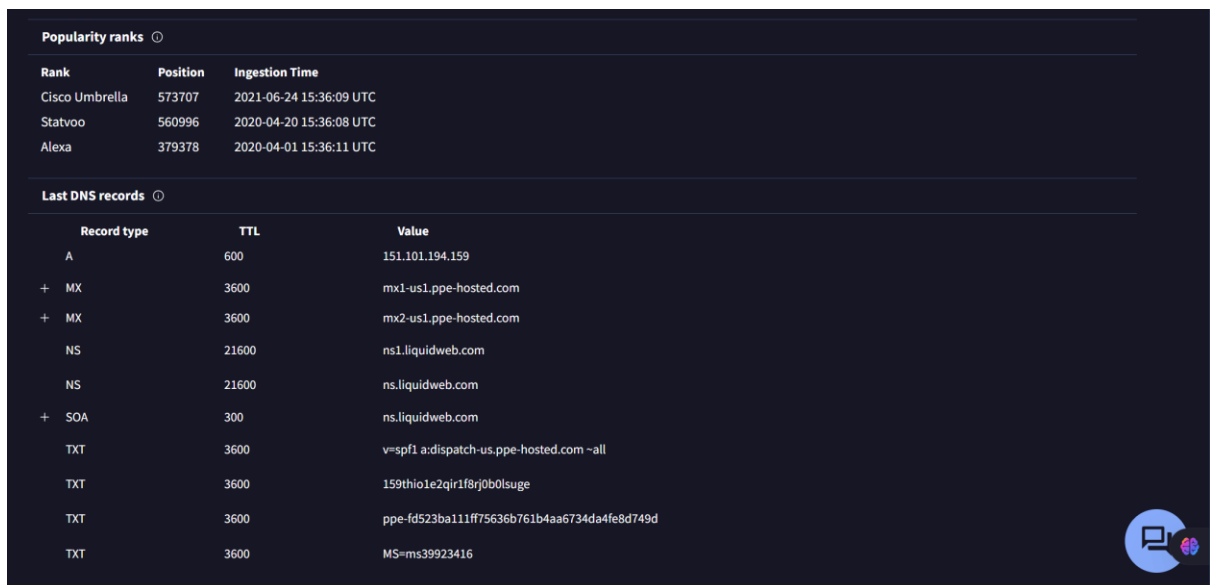
Domain Analyzed: maryhillmanor.org.



The screenshot shows the VirusShare interface for the domain maryhillmanor.org. The 'DETECTION' tab is active, displaying a table of security vendors' analysis. A banner at the top encourages joining the community. The table lists various vendors and their detection results, with a link to automate checks.

Security vendors' analysis		Do you want to automate checks?	
alphaMountain.ai	Malicious	BitDefender	Phishing
CRDF	Malicious	CyRadar	Malicious
G-Data	Phishing	Sophos	Malware
Abusix	Clean	Acronis	Clean
ADMINUSLabs	Clean	AILabs (MONITORAPP)	Clean
AlienVault	Clean	Antiy-AVL	Clean
benkow.cc	Clean	Blueliv	Clean

Fig.1 Security Vendor Flag List



The screenshot shows the VirusShare interface for the domain maryhillmanor.org. The 'DETECTION' tab is active, displaying a table of popularity ranks and a table of last DNS records.

Popularity ranks		
Rank	Position	Ingestion Time
Cisco Umbrella	573707	2021-06-24 15:36:09 UTC
Statvoo	560996	2020-04-20 15:36:08 UTC
Alexa	379378	2020-04-01 15:36:11 UTC

Last DNS records		
Record type	TTL	Value
A	600	151.101.194.159
+ MX	3600	mx1-us1.ppe-hosted.com
+ MX	3600	mx2-us1.ppe-hosted.com
NS	21600	ns1.liquidweb.com
NS	21600	ns.liquidweb.com
+ SOA	300	ns.liquidweb.com
TXT	3600	v=spf1 a:dispatch-us.ppe-hosted.com ~all
TXT	3600	159thio1e2qir1f8rj0b0lsuge
TXT	3600	ppe-fd523ba111ff75636b761b4aa6734da4fe8d749d
TXT	3600	MS=ms39923416

Fig. 2 Security Vendor Analysis List

File Analyzed: mysql-installer-community-8.0.42.0.msi

- **Detection Results:** 0/70 antivirus engines flagged it as malicious, considered safe.
- **Metadata:** The file type is msi, associated with installation files.
- **Behavioural Checks:** No suspicious activity detected.

Basic properties	
MD5	b0406f4ea3e5942909f6b054f9575e12
SHA-1	fc25864c1c84dea25bf7676bbfc2ef74c034e62
SHA-256	04ac71a3f67c686b0745beebf5edd317eca0973285c4970fbd0aeadd0c5a4c86
Vhash	753077f41932eb14e257346f70a52582
SSDEEP	6291456:wJIX53Ryb0kO2AFTI5joj7DrItmH/ul2grolQvumfAq2CEIMxa9wu8y2:XJ3j5ZfipH/uToev/+u9/2
TLSH	T12C093394F8A20E24F7FEC2348275AB14C3DB0E29F751400FE9BDDA9645B736456E920B
File type	Windows Installer installer windows msi
Magic	Composite Document File V2 Document, Little Endian, Os: Windows, Version 6.2, MSI Installer, Code page: 1252, Title: Installation Database, Subject...
TrID	Microsoft Windows Installer (83.7%) Windows SDK Setup Transform script (11.2%) Kingsoft WPS Office document (alt.) (3.5%) Generic OLE...
File size	353.66 MB (370843648 bytes)
History	
Creation Time	2025-04-09 07:00:52 UTC
Signature Date	2025-04-09 07:05:00 UTC
First Submission	2025-04-16 20:42:31 UTC
Last Submission	2025-08-20 14:57:11 UTC
Last Analysis	2025-08-20 09:40:27 UTC
Names	
mysql-installer-community-8.0.42.0.msi	
mysql-installer-community-8.0.42.0 (1).msi	
mysql-installer-community-8.0.42.0 (2).msi	
mysql-installer-community-8.0.42.0(1).msi	

Fig. 3 mysql-installer-community-8.0.42.0.msi file check

Result:

The experiment successfully confirmed that the domain maryhillmanor.org and the file **mysql-installer-community-8.0.42.0.msi** are both safe. The domain showed high credibility across multiple sources, and the file was not flagged as malicious by any of the antivirus engines, confirming it is safe for use.

Precautions:

1. Do not visit suspicious domains directly in your browser.
2. Use VirusTotal only for analysis, not for browsing or interacting with harmful websites.
3. Double-check the spelling of the domain to avoid analyzing a fake site.

EXPERIMENT-2

Aim: Shodan's Explore page

Material Required:

4. Internet-connected computer
5. Shodan.com

Theory:

Shodan is a specialized search engine that scans the internet for connected devices and services. Unlike traditional search engines that index websites, Shodan collects data on IoT devices, servers, routers, webcams, and other internet-exposed hardware. It gathers information such as open ports, service banners, software versions, and vulnerabilities. Security professionals, researchers, and network administrators use Shodan to identify exposed systems, assess risks, and monitor their networks. By revealing the “hidden” internet infrastructure, Shodan helps improve cybersecurity awareness and response. It's a powerful tool for understanding the landscape of devices accessible over the internet worldwide.

Procedure

1. **Open Shodan website:**
 1. Go to <https://www.shodan.io>
2. **Log in or create an account (optional but recommended):**
 1. Sign up for a free account to get more detailed results.
 2. Log in if you already have an account.
3. **Enter the IP address in the search bar:**
 1. Type `188.246.6.199` and press **Enter**.
4. **Review the search results:**
 1. Check open ports on the device.
 2. Identify running services (e.g., HTTP, FTP).
 3. Examine service banners for software info.
 4. See geolocation and hostname details.
 5. Look for any known vulnerabilities (CVEs).
5. **Use filters if needed (optional):**
 1. Example: `port:80 188.246.6.199` to see only HTTP services.
 2. Example: `net:188.246.6.0/24` to check the whole subnet.
6. **Analyze the data:**
 1. Assess which services are exposed.
 2. Note software versions for vulnerabilities.
 3. Check scan timestamps for freshness.

OBSERVATIONS

Shodan Port Scanning

SHODAN

Explore

Downloads

Pricing

scan

Q

Account

TOTAL RESULTS

37,596

TOP COUNTRIES

Germany	11,963
United States	8,743
France	3,245
Canada	1,700
Taiwan	1,323
More...	

TOP PORTS

993	25,660
143	6,358
8889	640

Product Spotlight: Free, Fast IP Lookups for Open Ports and Vulnerabilities using InternetDB

103.17.8.66

103-17-8-66.static.ip.net.tw
Yuan-Jhen Info., Co., Ltd
Taiwan, Banqiao

188.246.6.199

mail.arneschrader.de
nextcloud.moestl.name
nextcloud.arneschrader.de
nuncents.moestl.name
Portunity GmbH
Germany, Forst

92.112.60.217

Cactus Hospedagem LTDA

SSL Certificate

Issued By:
Common Name:
R10
Organization:
Let's Encrypt
Issued To:
Common Name:
nextcloud.arneschrader.de
Supported SSL Versions:
TLSv1, TLSv1.1, TLSv1.2

* OK [CAPABILITY IMAP4rev1 LITERAL+ ID ENABLE AUTH-PLAIN AUTH-LOGIN SASL-I

* CAPABILITY IMAP4rev1 LITERAL+ ID ENABLE ACL RIGHTS-kxten QUOTA MAILBOX-R

SSL Certificate

* OK [CAPABILITY IMAP4rev1 LITERAL+ ID ENABLE AUTH-PLAIN SASL-IR] example.

SHODAN

Explore

Downloads

Pricing

Search

Q

Account

188.246.6.199

Regular View

Raw Data

Timeline

OpenMapTiles Satellite

MapTiler

OpenStreetMap contributors

// TAGS: starttls

// LAST SEEN: 2025-08-28

General Information

mail.arneschrader.de

nextcloud.arneschrader.de

nextcloud.moestl.name

nuncents.moestl.name

Domains

arneschrader.de

moestl.name

Country

Germany

City

Forst

Organization

Portunity GmbH

ISP

Portunity GmbH

ASN

AS15987

Open Ports

25

80

143

443

465

587

993

// 25 / TCP

983113481 | 2025-08-23T09:28:20.034669

Postfix smtpd

220 nuncents.moestl.name ESMTP Postfix
250-nuncents.moestl.name
250-PIPELINING
250-SIZE 64000000
250-VRFY
250-ETRN
250-STARTTLS
250-ENHANCEDSTATUSCODES
250-BBIMIME
250-DSN

SSL Certificate

Domains

arneschrader.demoestli.name

Country

Germany

City

Forst

Organization

Portunity GmbH

ISP

Portunity GmbH

ASN

AS15987

Web Technologies

Security

HSTS

Web Servers

Apache HTTP Server

UI Frameworks

Bootstrap

Postfix smtpd

220 muncent.s.moestli.name ESMTP Postfix
250-muncent.s.moestli.name
250-PIPELINING
250-SIZE 64000000
250-VRFY
250-ETRN
250-STARTTLS
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN

SSL Certificate

Certificate:
Data:
Version: 3 (0x2)
Serial Number:
05:be:ab:a1:6d:44:97:1e:0f:a0:cf:87:b7:20:2a:76:e7:23
Signature Algorithm: sha256WithRSAEncryption
Issuer: C=US, O=Let's Encrypt, CN=R11
Validity
Not Before: Jul 19 23:06:44 2025 GMT
Not After : Oct 17 23:06:43 2025 GMT
Subject: CN=nextcloud.arneschrader.de
Subject Public Key Info:
Public Key Algorithm: rsaEncryption
Public-Key: (4096 bit)
Modulus:
00:b8:ae:a8:f3:47:5b:e9:8c:5a:9e:e7:dd:77:bd:
ce:04:9d:96:1e:11:bfc:ea:6:81:84:75:d6:f7:b8:
82:d5:4d:cb:7a:db:12:b2:e9:32:70:a4:a2:01:db:
58:ac:a5:92:71:22:04:77:22:57:88:77:ec:de:8e:
82:1a:c8:94:20:6a:0f:0f:89:eb:87:6a:e9:8a:1c:
4e:23:f2:f4:a1:eb:62:17:76:b0:44:01:0b:83:74:
82:b5:f6:14:a3:53:ea:cf:f3:99:a8:c2:dd:57:92:
45:c3:33:3d:e5:de:48:12:e0:9a:ba:54:b2:95:41:

Result:

The Shodan search revealed open ports, detected running services, and displayed software banner information for the target IP. Geolocation and host details were identified, highlighting potential vulnerabilities. This analysis demonstrates Shodan’s capability to assess exposed devices and network risks effectively.

EXPERIMENT-3

Aim: To use Recuva software to restore deleted files on a laptop/PC.

Material Required:

- A laptop or PC with Windows OS
- Recuva software (Free version is sufficient)
- Internet connection (for downloading the software, if not already installed)
- A file or folder that has been recently deleted (for testing)

Theory:

Recuva is a file recovery tool developed by Piriform that allows users to recover files accidentally deleted from their computer, including files emptied from the Recycle Bin. It can also recover files from external drives, memory cards, USB sticks, and other storage devices. Recuva works by scanning sectors of the drive for data remnants that have not yet been overwritten. It offers both a quick scan and a deep scan mode for locating recoverable files. This software is widely used in data recovery, digital forensics, and IT troubleshooting.

Procedure

• Download and Install Recuva:

1. Go to the official website: <https://www.ccleaner.com/recuva>
2. Download the Recuva installer and follow the setup instructions.

• Open Recuva:

3. Launch Recuva from the Start Menu or desktop shortcut.

• Start the Recuva Wizard:

4. Click **Next** on the Recuva Wizard welcome screen.
5. Choose the type of file you want to recover (e.g., Pictures, Documents, Videos, or All Files).
6. Click **Next**.

• Select File Location:

7. Choose where the file was deleted from:
 1. In the Recycle Bin
 2. In a specific location
 3. On a memory card or USB drive
 4. Don't know (to scan the entire drive)
8. Click **Next**.

- **Enable Deep Scan (Optional):**

1. If the file was deleted a long time ago or the quick scan does not find it, check the box for **Enable Deep Scan**.
2. Click **Start** to begin scanning.

- **Review the Scan Results:**

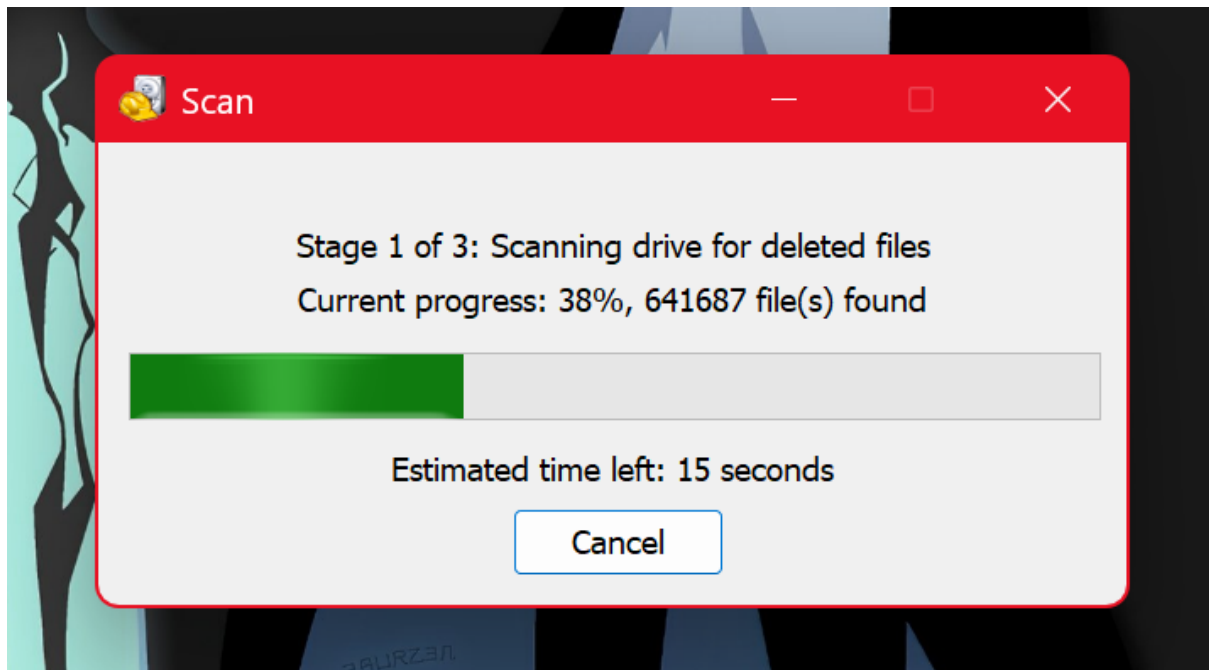
1. Wait for the scan to complete.
2. Recuva will display a list of recoverable files with a color-coded status:
 1. **Green:** Excellent (high chance of full recovery)
 2. **Orange:** Poor (partially recoverable)
 3. **Red:** Unrecoverable

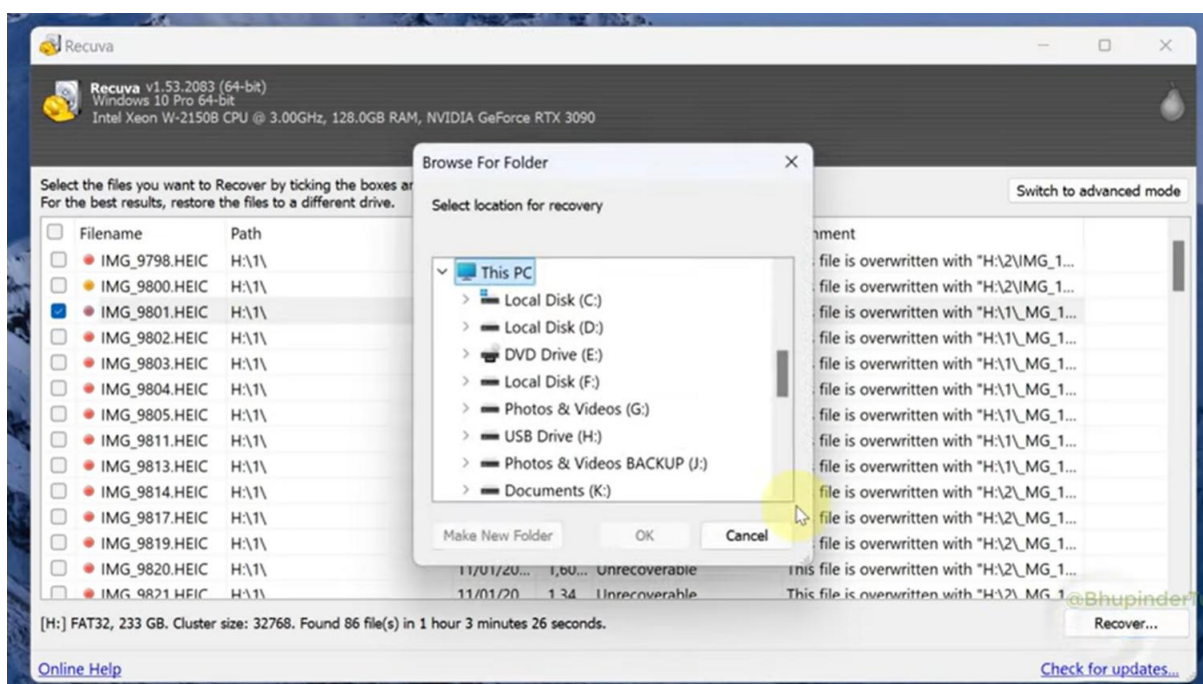
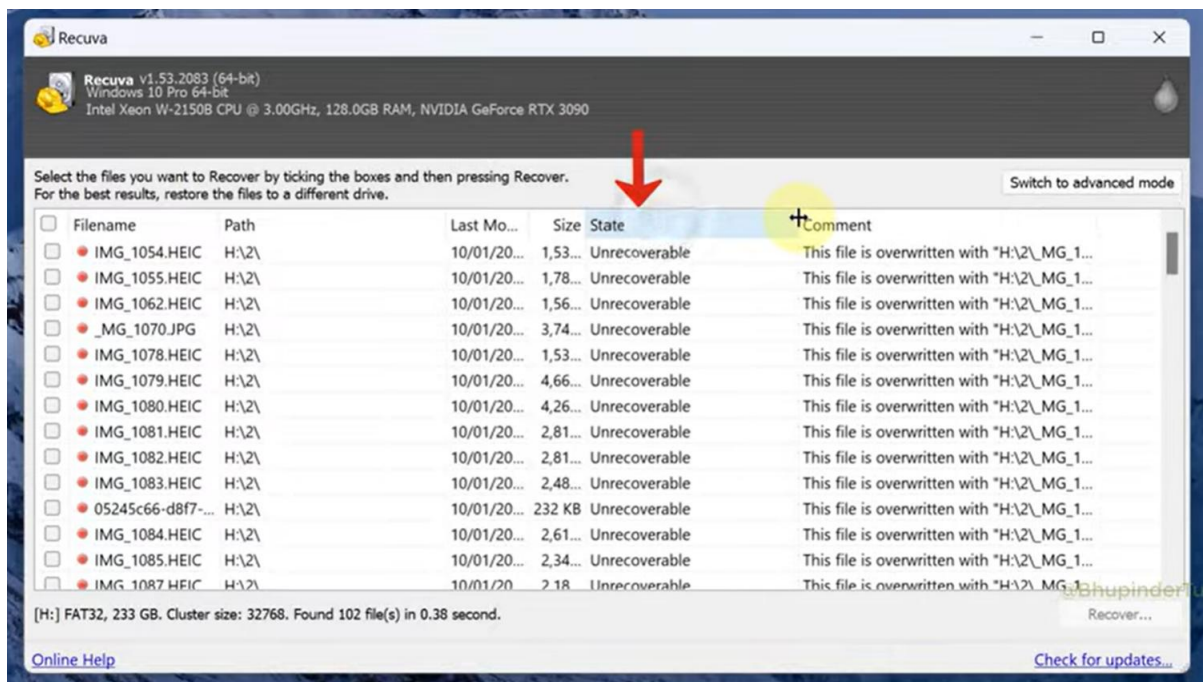
- **Recover the Files:**

1. Select the file(s) you wish to recover by checking the box.
2. Click **Recover** and choose a different drive or folder to save the recovered files (to avoid overwriting them).

OBSERVATIONS

Shodan Port Scanning





Result: Recuva successfully identified and restored deleted files from the system. Files with a green status were fully recovered, while some with a red status were unrecoverable due to data being overwritten. This experiment demonstrated Recuva's efficiency in recovering lost data and emphasized the importance of acting quickly after file deletion to maximize recovery success.

EXPERIMENT-3

Aim: Finding Hardcoded Credentials from local hosts lists

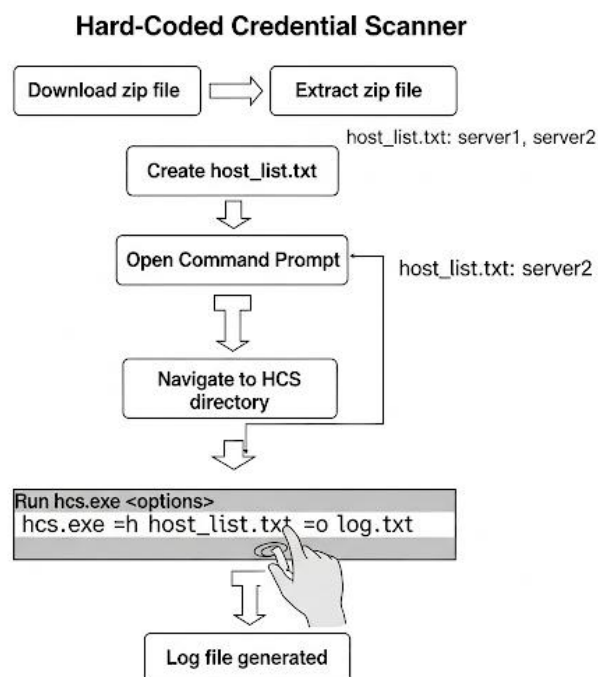
Material Required:

3. Hcs.exe from Bruno Caseiro
4. Host_list.txt
5. Windows computer

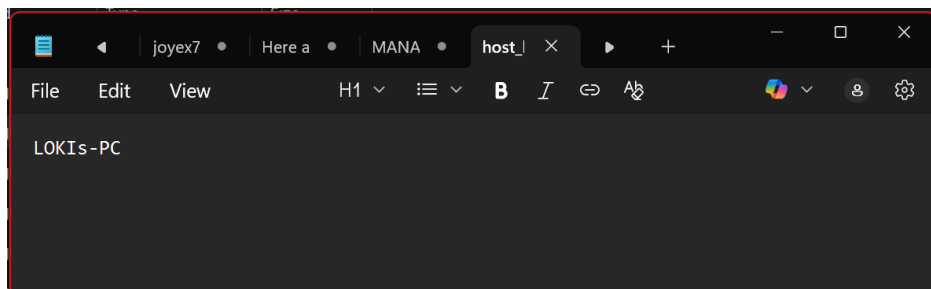
Theory:

The HCS (Hard-Coded Credential Scanner) is a security tool designed to identify hard-coded credentials within specific file extensions across a list of systems. This experiment aims to assess its effectiveness in detecting security vulnerabilities related to embedded credentials in codebases. By deploying HCS across a network of systems, the experiment will evaluate its ability to scan designated file types for hard-coded credentials. The tool's efficiency will be measured by its accuracy in identifying such vulnerabilities and generating comprehensive logs for analysis. This assessment is crucial for understanding the tool's applicability in real-world security audits and its potential integration into continuous security monitoring processes.

Procedure:



Observations:



```
Command Prompt
ers that you want to audit. *
* Detailed output will be generated in a log file named HCS.LOG, including all findings.
*
* Note: Hidden shares such as C$, Admin$, etc are not going to be scanned.
*
*****
*****
* Author: Bruno Caseiro -- bacaseiro@gmail.com | Use at your own risk
*****
*****
2025-09-04 11:35 ==> Starting the process of looking for hardcoded credentials in .TXT, .INI, .CONFIG, and .LOG files.
2025-09-04 11:35 ==> Connecting on the system: LOKIs-PC
2025-09-04 11:35 ==> Checking for open shares on the host..: LOKIs-PC

-----
--
-- FINDINGS SUMMARY:
--
-----

2025-09-04 11:35 ==> Scan has been completed. Time taken to run this scan: 0:00:00

C:\Users\Psn\Downloads\hcs>
```

Result:

The HCS tool effectively detects hard-coded credentials in specified file types, enhancing vulnerability identification. Its scans generate detailed reports, aiding in mitigating security risks and improving codebase integrity during security audits.

Precautions:

6. Ensure scanning only targets relevant file extensions to avoid false positives.
7. Backup codebases before scanning to prevent accidental data loss.
8. Regularly update the tool for latest vulnerability patterns.
9. Limit access to scan results to authorized personnel only.

Experiment 5

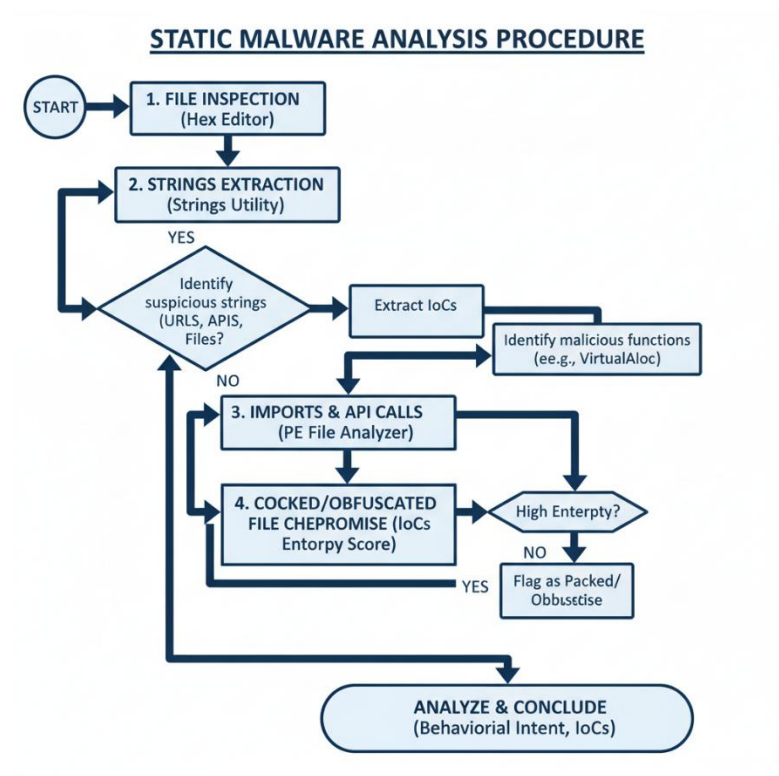
Aim: To perform Static Analysis of a Malware

Materials Required: Windows/Linux System, Hex Editor, Strings Utility, PE File Analyzer

Theory: Static Analysis is the process of examining a suspicious file or software without executing it. This method helps identify malicious behaviors, patterns, or structures by analyzing the file's code, headers, strings, and metadata. It is a safe and efficient way to understand the characteristics of a file without risking infection or damage.

10. File Inspection:
Tools like a hex editor are used to view the file's structure and confirm its format by identifying magic numbers (e.g., MZ for Windows executables).
11. Strings Extraction:
Analysts search for readable text within the file such as URLs, filenames, API calls, and registry paths that may point to malicious activity.
12. Imports and API Calls:
Examining imported libraries and functions reveals potential threats. Functions like CreateRemoteThread or VirtualAlloc are commonly used by malware to inject code or manipulate memory.
13. Packed or Obfuscated Files:
Files may be compressed or encrypted to hide malicious code. High entropy scores can indicate such obfuscation.
14. Indicators of Compromise (IoCs):
URLs, DLL names, registry paths, and suspicious APIs help determine whether the file behaves maliciously.

Procedure:



Observations:

Mock: Hex Editor View (showing MZ header)

```
HxD - singleFileHost.exe    Offset: 00000000h
00000000  4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 MZ.....
00000001  4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 MZ.....@.....
00000002  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00000003  2E 74 78 74 00 00 00 00 00 00 00 00 00 00 00 00 txt.....
00000004  50 45 00 00 4C 01 03 00 00 00 00 00 00 00 00 00 PE..L.....
```

Mock: Strings Extractor Output (ranked)

```
Stringsifter - Ranked strings from singleFileHost.exe

Rank | String (excerpt)                                     |
-----|-----|
1.   | http://malicious.example/payload                     | URL
2.   | CreateRemoteThread                                   | API
3.   | singleFileHost                                       | Filename
4.   | suspicious.dll                                       | DLL
5.   | HKEY_LOCAL_MACHINE\Software\...                      | Registry
```

Mock: PE Analyzer Summary (suspicious indicators)

Pestudio Basic – Analysis Summary

File: singleFileHost.exe

Size: 1,234,560 bytes

MD5: d41d8cd98f00b204e9800998ecf8427e

Suspect imports: CreateRemoteThread, VirtualAlloc, LoadLibraryA

Packed/Obfuscated: Likely (high entropy)

Suspicious strings: http://malicious.example, CreateRemoteThread

Score: High (75/100)

Result:

Successfully performed static analysis. Inspected file headers, extracted meaningful strings like URLs and API calls, and analyzed imported functions. Identified potential **Indicators of Compromise (IoCs)**, forming a preliminary threat assessment without executing the code.

Precautions:

1. Always perform analysis in an **isolated virtual machine (VM)** to contain any potential accidental execution or side effects.
2. **Disable all network connections** on the analysis machine to prevent the malware from contacting external command-and-control servers.
3. Never double-click or directly execute the malware sample; only open it with the **specified static analysis tools**.

EXPERIMENT-6

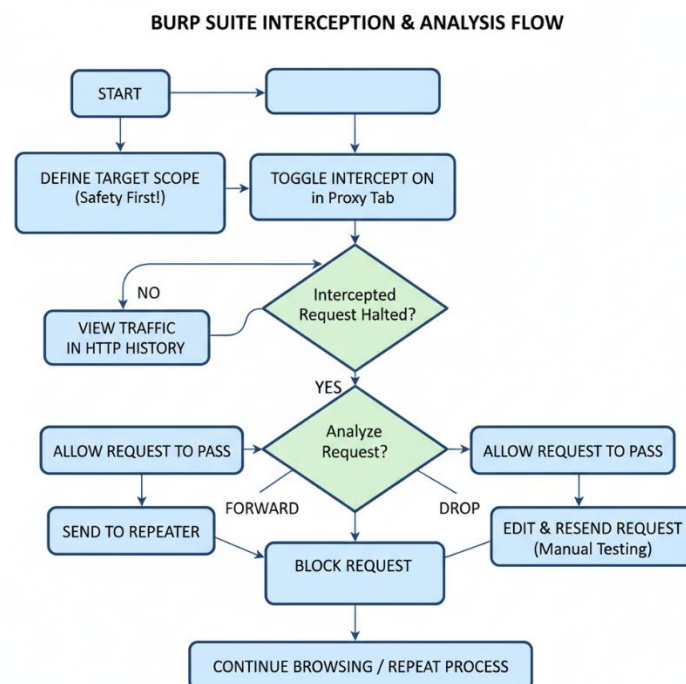
Aim: To intercept, analyse, and manipulate HTTP/HTTPS web traffic via Burp Suite, thereby identifying web application vulnerabilities.

Material Required: Computer System, Internet Connection, Web Browser, Burp Suite Software, Target Web Application

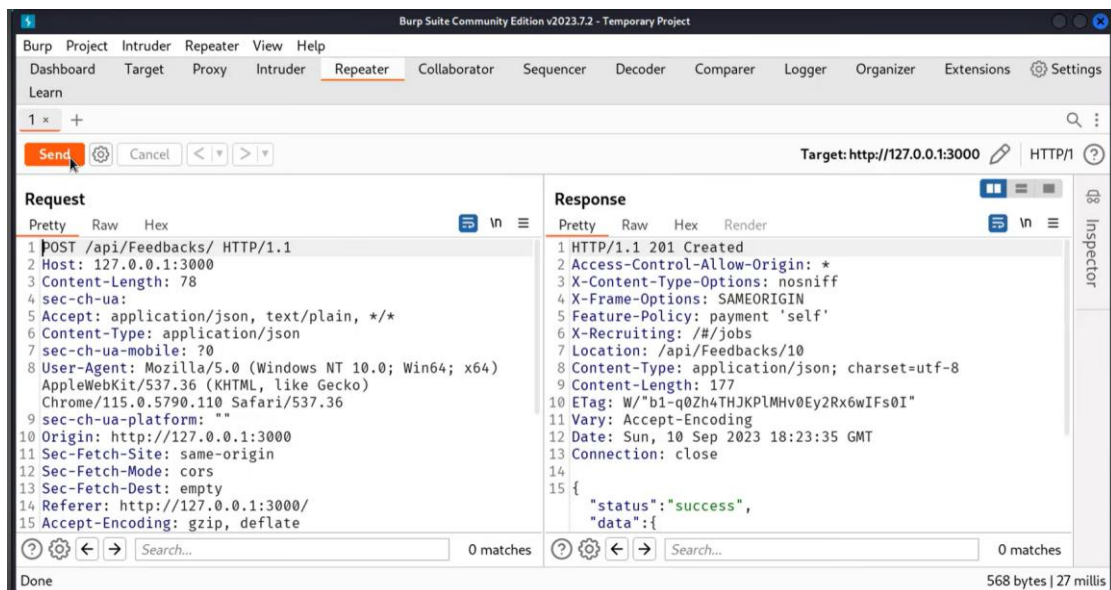
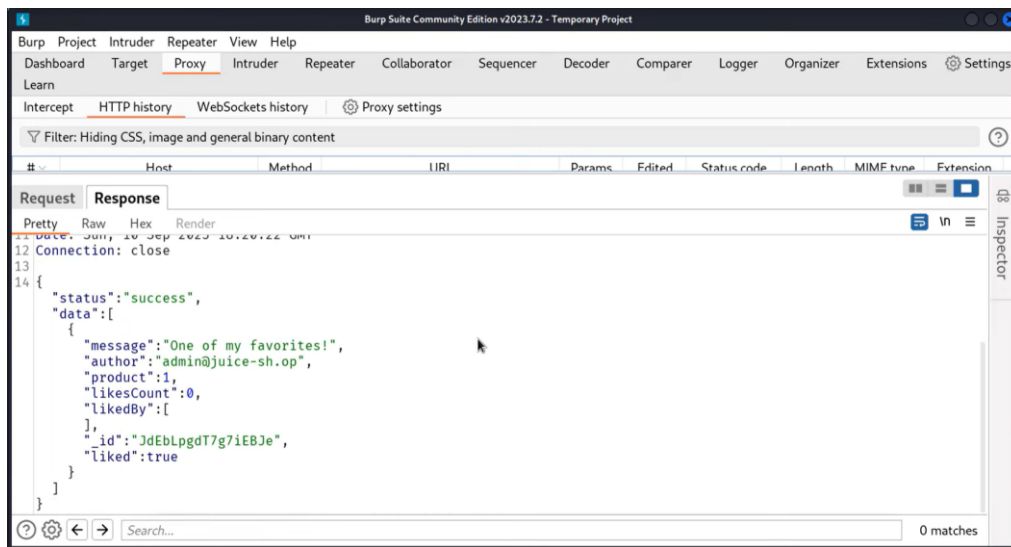
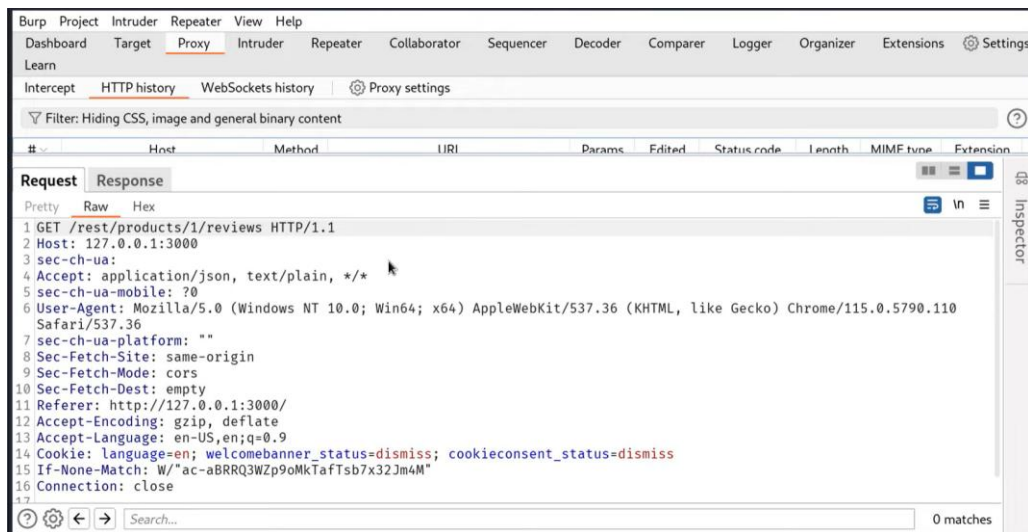
Theory:

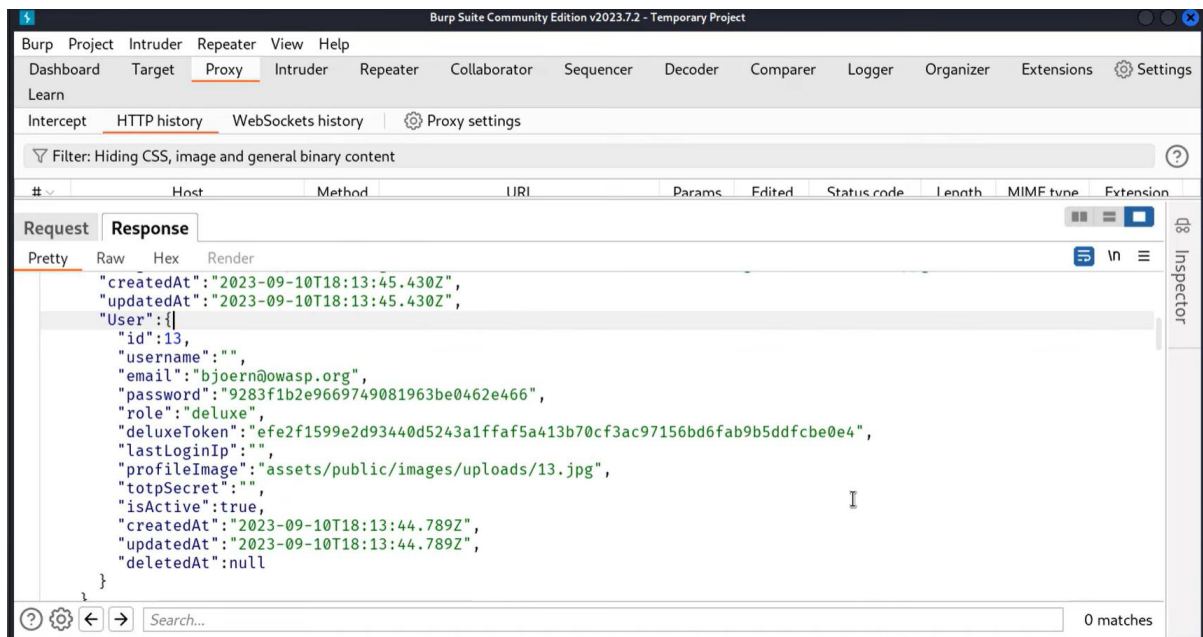
Web application security relies on analyzing client-server communication. A web proxy, such as Burp Suite, operates as a "man-in-the-middle," intercepting all traffic between a user's browser and a web server. This allows security testers to view, modify, and replay HTTP/HTTPS requests and responses in real-time. By manipulating data sent to the server, one can probe for vulnerabilities like SQL injection, cross-site scripting (XSS), and improper access control. For encrypted HTTPS traffic, installing a custom CA certificate is necessary to allow Burp Suite to decrypt and re-encrypt the data for analysis.

Procedure:



Observations:





Results:

Successfully intercepted and analysed HTTP/HTTPS traffic. Understood web proxy function, modified requests, identified server responses, and learned basic web vulnerability testing. Gained practical web security insight.

Precautions:

4. Always obtain explicit permission before testing any web application. Unauthorized testing has legal consequences.
5. Strictly define and adhere to the target scope within Burp Suite to prevent unintended interactions with other sites.
6. Remove Burp Suite's CA certificate after use, especially on shared computers, to maintain browser security.
7. Be cautious with data modifications; ensure testing occurs in controlled, non-production environments to prevent harm.